

Studies on the phytoomodulatory potential of fenugreek (*Trigonella foenum-graecum*) on bisphenol-A induced testicular damage in mice

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Abstract

Bisphenol A (BPA), an organic synthetic compound and endocrine disruptor, which majorly cause deleterious effects on male reproductive system. Fenugreek (*Trigonella foenum-graecum*), associated with Leguminosae family is used as a herbal medicine with potent antioxidant properties. The present study was aimed to scrutinise the preventative role of fenugreek seeds aqueous extract (FSEt) on BPA-induced testicular damage in mice. Study included four different groups of male Balb/c mice: control (C), fenugreek (FSEt), bisphenol A (BPA) and fenugreek + bisphenol A (FSEt + BPA). After two months of treatment, assessment of sperm parameters, antioxidant defence system, histopathological studies, germ cell count and gene expression of intrinsic apoptotic pathway were carried out. Administration of FSEt improved the damage caused by BPA as indicated by improved sperm parameters. FSEt-administered mice showed improvement in the histoarchitecture compared with BPA-administered animals. In addition, fenugreek treatment showed reduced levels of malondialdehyde and elevated levels of antioxidant enzymes. Expression studies of apoptotic markers revealed a significant decrease in the expression of Bcl-2 and significant increase in caspase-9 and caspase-3. However, FSEt restored the deleterious effects caused by BPA. The current findings plausibly might have promising protective role against BPA-induced testicular damage.

KEYWORDS

antioxidants, apoptosis, bisphenol A, fenugreek, oxidative stress

1 | INTRODUCTION

Spermatogenesis is a process by which spermatozoa are produced by the process of mitosis and meiosis. Mammalian spermatogenesis is a network of developmental process to which a specialised form of spermatozoa is formed by the transformation of diploid progenitor germ cell (Ibtisham et al., 2017). However, any disruption in the normal process either partial or complete results in the decline of sperm concentration, thereby, leading infertility. As per the statistical data, infertility approximately affects 15% of the couple of reproductive

age and male factor is the prominent cause in almost 50% of the cases (Ajina et al., 2017). Further, it has been known to cause depression, social isolation, sexual dysfunction and anxiety (Arambula, Jima, & Patisaul, 2018). Worldwide, half of the cases of infertility have been attributed to abnormal spermatogenesis in which impaired sperm motility and concentration are deemed as the prime cause of infertility (Gandhi et al., 2017). Essentially, factors identified in the case of male infertility are believed to be physiological, genetic, lifestyle and environmental factors (Vecoli, Montano, & Andreassi, 2016). Environmental factors include exposure to toxicants such as heavy metals, organic polychlorinated dibenzodioxins, dicarboximide

fungicides and environmental phenols (Roychoudhury, Saha, & Saha, 2019).

Bisphenol A (BPA) is a synthetic oestrogen and an environmental toxicant which is present in innumerable consumer products and considered as highest utilising industrial chemicals worldwide (Alboghobeish et al., 2019). Humans and wildlife are widely and critically exposed to BPA due to its extensive use for the production of water pipes, drinking containers, dental monomers, medical equipments, etc (Kaur, Saluja, & Bansal, 2018). Worldwide 400 million tonnes of plastics produces annually and roughly 3.5 million tons was manufactured in 2008 (Goncalves, Semperebon, Biazzi, Mantovani, & Fernandes, 2018). Due to advancement in industrial and manufacturing activities, BPA is becoming ubiquitous in our living environment and have raised various public health concerns (Wu et al., 2018). Animal-based studies have reported that BPA exposure exerts undesirable effects on male reproductive system by altering the expression of genes associated with spermatogenesis and endocrine disruption (Kaur et al., 2018). Another study by Shirani, Alizadeh, Mahdavinia, and Dehghani (2019), showed a marked reduction in the levels of reduced glutathione and catalase whereas elevation in mitochondrial reactive oxygen species (ROS) and lipid peroxidation (LPO) levels along with mitochondrial membrane damage was observed upon BPA administration in rats.

As a well-versed phenomenon oxidative stress is an outcome of peroxidation and cellular oxidation of macromolecules suchlike lipids, carbohydrates, proteins and nucleic acids (Qiu et al., 2016). Many enzymes and transcription factors can be modulated by oxidative stress through a several signalling cascades (Baxter, Mittler, & Suzuki, 2013). It also plays an important role in the pathogenesis of several diseases by activating numerous signalling pathways. A report by Mahdavinia et al. (2019) suggested that oxidative stress is a result of increased ROS upon BPA exposure. Being extremely reactive, ROS can negatively affect the functions of many enzymes and transcription factors through a multitude of signalling cascades. Ultimately, these signals result in changes in gene expression, which influence the ability of the cell survival and apoptosis (Khalaf, Ahmed, Moselhy, Abdel-Halim, & Ibrahim, 2019). It has been also delineated that apoptosis of germ cells as well as generation of oxidative stress is a result of ROS generation by BPA which further compromises the capacity of leydig cells to be steroidogenic and ability to differentiate the germinal epithelium into normal spermatozoa (Hales et al., 2005; Skinner, Manikkam, & Guerrero-Bosagna, 2010).

Considering the aforementioned concerns, numerous nutritional antioxidants can be examined for its feasible protective character as an antioxidant against oxidative stress induced by BPA and apoptosis, such as vitamin A, vitamin E, beta-carotene, phytochemicals (lycopene, quercetin, niacin) and dietary condiments (cloves, cumin seeds, cardamom and fenugreek). In food preparations due to the strong flavour and aroma, fenugreek is frequently utilised as a spice (Liu, Kakani, & Nair, 2012). Fenugreek seeds have shown various curative qualities suchlike hypocholesterolaemic, antibacterial, antioxidant, antidiabetic agent, hepatoprotective effect and anticancer (Wani & Kumar, 2018). Neha, Anand, and Sunanda (2019) reported

that fenugreek seed extract showed a protective effect against glibenclamide-induced lipid peroxidation in hepatic tissue. Another study by Hamza, Elwy, and Badawi (2016) advocated that the fenugreek seeds (aqueous extract) appreciatively attenuate the testicular damage induced by cisplatin.

Therefore, taking into consideration the antioxidant potential of fenugreek in various cellular systems, the current study was planned with an aim to explore the protective role of fenugreek seeds (aqueous extract) against BPA-induced oxidative stress and apoptosis in mice testis.

2 | MATERIAL AND METHODS

Male mice (Balb/c) weighing 20–25 g were procured from the Central Animal House, Panjab University, Chandigarh (India), and housed in a standard animal facility under controlled temperature and water provided and 12/12-hr light/dark cycle with standard food and water provided ad libitum. The study was conducted and duly approved as per the accordance of Institutional Animals Ethics Committee.

2.1 | Preparation of fenugreek seeds aqueous extract (FSEt)

Fresh fenugreek seeds were purchased from the market and were washed with distilled water to get out of irrelevant matter, air-dried and grinded into a fine granulate using electrical grinder. Then, the fine granulate was mixed for 3 days (1 g of seed granulate in 100 ml) with distilled water; after that, centrifugation at 800 g was done and the resultant supernatant was withdrawn and lyophilised. This lyophilised sample was homogenised in distilled water followed by administration to mice through oral gavage at a dose level 200 mg/kg body weight.

2.2 | Qualitative analysis of Phytochemical in FSEt

Qualitative analysis of phytochemicals viz alkaloids, tannins, flavonoids, proteins, steroids/polysterols, glycosides and carbohydrates were carried out in FSEt by the method reported earlier Mir, Sawhney, and Jassal (2013). For the preparation of test solution, 100-mg fenugreek powder was dissolved in 10 ml of distilled water.

2.2.1 | Alkaloids

Dragendorff's and Wagner's reagents were used to analyse the presence of alkaloids.

1. Dragendorff's reagent: 14 g of sodium iodide with 5.2 g of basic bismuth carbonate was added to 50 ml glacial acetic acid and boiled for a few minutes. It was then allowed to stand overnight

and precipitate of sodium acetate crystals were filtered off. To 40 ml of red-brown filtrate, 160 ml of ethyl acetate and 1 ml of distilled water was added (stock solution). The stock solution was preserved in an amber colour bottle. To prepare the working solution, add 20 ml of acetic acid to 10 ml of stock solution and make up the volume to 100 ml. 2–3 drops of Dragendorff's reagent was added to 2 ml of test solution. Appearance of brown precipitate indicated the presence of alkaloids.

2. Wagner's reagent: 1.27 g of iodine and 2 g of potassium iodide were dissolved in 100 ml of distilled water. 2–3 drops of Wagner's reagent was added to 2 ml of test solution. Appearance of reddish brown or green colour showed the presence of alkaloids.

2.2.2 | Tannins

Ferric chloride and lead acetate tests were used to analyse the presence of tannins.

1. Ferric chloride solution: 5% ferric chloride [FeCl₃] (w/v) in 90% ethanol. 0.2 ml of FeCl₃ was added to 2 ml of test solution. Appearance of black colour indicated the presence of tannins.
2. Lead acetate solution: 25% lead acetate in distilled water 0.3 ml of lead acetate solution was added to 2 ml of test solution. Appearance of white precipitate showed the presence of tannins.

2.2.3 | Flavonoids

Shinoda test was used to analyse the presence of flavonoids.

1. Shinoda test: The dried extract was dissolved in 95% ethanol, and few drops of concentrated hydrochloric acid were added. Then, the magnesium turning was added into the solution. Appearance of pink colour indicated the presence of flavonoids.

2.2.4 | Proteins

Millon's and ninhydrin reagents were used to analyse the presence of proteins.

1. Millon's reagent: 1 g of mercury was dissolved in 9 ml of fuming nitric acid under cooling. To this, an equal amount of distilled water was added. Two millilitre of Millon's reagent was added to 2 ml of test solution. Appearance of white precipitate indicated the presence of proteins.
2. Ninhydrin reagent: Dissolve 0.5 g of ninhydrin and 1 ml of 0.01 M potassium cyanide into 59 ml of methyl cellosolve. Two millilitre of Ninhydrin reagent was added to 2 ml of test solution. Appearance of white precipitate showed the presence of proteins.

2.2.5 | Steroids/polysteriods

Liebermann–Burchard reagent was used to analyse the presence of steroids/polysteriods.

1. Liebermann–Burchard reagent: Ten millilitre of acetic anhydride was mixed with 10 ml of concentrated sulphuric acid under cooling. To this, 100 ml of ethanol was added under cooling. Two millilitre of Liebermann–Burchard reagent was added to 2 ml of test solution. Appearance of green colour showed the presence of steroidal saponins.

2.2.6 | Glycosides (Anthraquinone glycosides and Cardiac glycosides)

Bontrager's, sodium picrate and legal tests were used to analyse the presence of glycosides.

1. Bontrager's test: Diluted ammonia solution was added to the test solution and shook for a minute. Appearance of pink/red/violet colour indicated the presence of anthraquinone glycosides.
2. Legal test: To 2 ml of test sample, 1 ml of pyridine solution and few drops of freshly prepared 2–3 drops sodium nitroprusside solution were added. Appearance of white colour showed the presence of cardiac glycosides.
3. Sodium picrate test: 2–3 drops of sodium picrate solution was added to 1 ml of test solution. Change in colour indicated the presence of cardiac glycosides.

2.2.7 | Carbohydrates

Fehling and Benedict's tests were used to analyse the presence of carbohydrates.

1. Fehling's solution: 34.6 g of copper sulphate was dissolved in water, and final volume was made to 500 ml (Solution A) using distilled water. 173 g of potassium sodium tartrate and 50 g of sodium hydroxide were dissolved in distilled water, and the final volume was made to 500 ml (solution B) using distilled water. The two solutions were mixed in equal volume prior to use. Fehling solution was added to the test solution and boiled for 2–3 min. Appearance of red colour indicated the presence of carbohydrates.
2. Benedict's solution: One litre of benedict's solution was prepared by dissolving 100 g of anhydrous sodium carbonate, 173 g of sodium citrate and 17.3 g of copper sulphate pentahydrate in a total volume of 1,000 ml. 2–3 drops of Benedict's solution was added to 1 ml of test solution. Appearance of red colour showed the presence of carbohydrates.

2.3 | Treatment schedule

Animals were segregated into four different groups. Group I animals fed with normal pellet diet and water ad libitum for 8 weeks served as normal control Group I (C). Group II animals supplemented with fenugreek seeds aqueous extract by oral gavage at a dose level 200 mg/kg body weight (Yadav et al., 2010) served as fenugreek seeds aqueous extract-treated Group (FSEt). Group III animals administered with BPA (1 mg/kg body weight) (Goncalves et al., 2018) by oral gavage served as bisphenol A Group (BPA). Group IV animals co-supplemented with FSEt and BPA by oral gavage served as fenugreek + bisphenol A Group (FSEt + BPA).

2.4 | Sperm concentration and motility

2.4.1 | Sperm concentration

Spermatozoa were teased out from epididymis and suspended in 1 ml of 0.9% NaCl (normal saline). This suspension was then placed on haemocytometer, and the spermatozoa numbers were counted. The counted spermatozoa numbers were then indicated as million/ml (Goyal, Koul, & Ranawat, 2019).

2.4.2 | Sperm motility

From vas deferens, the spermatozoa were taken on a glass slide carrying 1 ml normal saline and mixed. This suspension (100 µl) was then placed on a clean glass slide and covered with a coverslip. In several random microscopic fields, the motile and nonmotile spermatozoa were counted and indicated as the percentage of the motile spermatozoa (Goyal et al., 2019).

2.5 | Light microscopy

Histopathological changes in various treatments groups were studied using light microscopy. After dissection testes were taken out and shifted into formalin for fixation. Testes were then placed in a graded series of alcohol for dehydration and fixed in paraffin wax. Thick sections of tissue were cut using microtome and stained with haematoxylin and eosin.

2.6 | Testicular germ cell count study

From haematoxylin-and-eosin-stained slides, random fields were selected for germ cells count. The germ cells were counted in the twenty-five randomly selected sections of seminiferous tubules (Leblond & Clermont, 1952).

2.7 | Biochemical estimations

The amount of reactive oxygen species (ROS), lipid peroxidation (LPO), reduced glutathione (GSH) and glutathione peroxidase (GSH-Px) determined the oxidant/antioxidant status of testicular tissue. To examine the levels of ROS, LPO and GSH, the method by Driver, Kodavanti, and Mundy (2000), Wills (1966), and Moron, Depierre, and Mannervik (1979) were used respectively. The GSH-Px activity was analysed by the technique of Lawrence and Burk (1976).

2.8 | Reverse transcription-polymerase chain reaction analysis

For reverse transcription-polymerase chain reaction analysis (RT-PCR) analysis of Bcl-2, caspase-9 and caspase-3, the total RNA was isolated from testes using tri-reagent. For loading control, β -Actin was used. The gene sequence of β -Actin (forward primer: 5'-ATCCGTAAAGACCTCTATGC-3'; reverse primer: 5'-AACGCAGCTCAGTAACAGTC-3'), Bcl-2 (forward primer: 5'-AGAGGGGCTACGAGTGGGAT-3'; reverse primer: 5'-CTCAGTCATCCACAGGGCGA-3'); caspase-9 (forward primer: 5'-TGCACCTCCTCTCAAGGCAGGACC-3'; reverse primer: 5'-TCCAAGGTCTCCATGTACCAGGAGC-3'); caspase-3 (forward primer 5'-AGTCAGTGGACTCTGGGATC-3'; reverse primer: 5'-GTACAGTTCTTCGTGAGCA-3'). By using RT-PCR kit, the reaction mixed was prepared and run in the thermal cycler. Agarose gel electrophoresis of PCR products was performed, and the densitometric analysis of the resultant bands was done by using ImageJ software (NIH, USA).

2.9 | Statistical analysis

The results were indicated as mean $\pm$ standard deviation ($n = 6$) of independent observations in each group for statistical significance. For the data analysis, one-way ANOVA was used followed by post hoc test for the comparison of means for different treatment groups using least significant difference (LSD). "a, b and c" represent comparison with respect to Group I (C); Group II (FSEt); and Group III (BPA). Here, 1, 2 and 3 represent $p < .001$, $p < .01$ and $p < .05$ respectively.

3 | RESULTS

3.1 | Phytochemical analysis of FSEt

The current study uncovers the medicinal active constituents present in fenugreek seeds. The qualitative analysis of active compounds present in FSEt was carried out, and the results are

shown in Table 1. In case of tannins, observed black colour indicates the presence of tannin. Similarly, change in colour was also observed in case of flavonoids and proteins whereas alkaloids, steroids, glycosides and carbohydrates gave negative observations.

3.2 | Organ weight

A significant reduction ($p < .001$) was found in the testis weight of Group III (BPA) when compared with Group I (C). However, a significant increase ($p < .01$) was found in the testis weight of Group IV (FSEt + BPA) when compared with Group III (BPA) which shows that fenugreek's administration compensates the loss generated by ROS to a much better extent in Group IV (Table 2).

3.3 | Sperm characteristics

Fenugreek seed extract prevented the bisphenol A-induced reduction in spermatozoa (Table 2).

In Group III (BPA), receiving BPA alone had significantly reduced ($p < .001$) sperm concentration to the corresponding Group I (C); however, a marked elevation ($p < .01$) in the sperm concentration was found in Group IV (FSEt + BPA) when compared with Group III (BPA), recommending the protective effect of FSEt against BPA-induced testicular toxicity.

Group III (BPA) had significantly lower ($p < .001$) sperm motility when compared to the Group I (C), but Group IV (FSEt + BPA) animals revealed noteworthy increase ($p < .001$) in sperm motility compared with Group III (BPA). The aforesaid findings indicate the defensive effect of FSEt upon BPA triggers testicular damage (Table 2).

3.4 | Histopathological studies

The histological analysis was performed to observe the architecture of testicular tissue (Figure 1). Testicular tissue of Group I (C) and Group II (FSEt) animals showed expected histoarchitecture and morphology (Figure 1a,b). Transformation of spermatogonia to mature spermatozoa was distinctly seen in the seminiferous tubules of mice testes in both the groups (Group I and Group II). However, high vacuolisation in germinal epithelium, irregularity in basement membrane and decline in number of spermatogonia, spermatocytes, round and elongated spermatids were found in Group III (BPA) (Figure 1c). In Group IV (FSEt + BPA), an improved histoarchitecture was observed and germ cell population was systematically arranged. Also, basement membrane was preserved with less vacuolisation and elongated, round spermatids number was increased as compared to Group III (BPA) (Figure 1d). This suggests that FSEt helps in normalising the histoarchitecture distorted by BPA administration.

TABLE 1 Phytochemical analysis of fenugreek seeds aqueous extract (FSEt)

Phytochemicals	Test	Inference
Alkaloids	Dragendorff's test	–
	Wagner's test	–
Tannins	Ferric chloride test	+
	Lead acetate test	+
Flavonoids	Shinoda test	+
Proteins	Millon's reagent	+
	Ninhydrin test	+
Steroids/polysterols	Liebermann–Burchard test	–
Glycosides	Bontrager's test	–
	Sodium picrate	–
Carbohydrates	Fehling's solution test	–
	Benedict's solution test	–

3.5 | Testicular germ cell number studies

Haematoxylin-and-eosin-stained sections were used to evaluate the germ cell number in various treatment groups. Number of various germ cells suchlike spermatogonia (SG), spermatocytes (SC), round spermatids (RSt) and elongated spermatids (Est) were quantified (Figure 2). In Group III (BPA) animals, SG, SC, RSt and Est number were significantly reduced as compared to control animals. However, animals of Group IV (FSEt + BPA) showed a remark elevation ($p \leq .001$) in these cells.

3.6 | Biochemical assays

To investigate the effect of fenugreek seeds aqueous extract on oxidative stress originate by BPA exposure in testicular tissue, assessment of peroxidative injury by measuring ROS levels (Figure 3) and LPO levels (Figure 4), activity of antioxidant enzymes such as GSH (Figure 5) and GSH-Px (Figure 6) were carried out. ROS and LPO were appreciably refurbished ($p \leq .001$) in case of Group IV (FSEt + BPA) animals which were found to be the escalated in case of Group III (BPA). Antioxidant enzymes (viz. GSH, GSH-Px,) which serve to maintain the redox balance of cells were found to be diminished ($p \leq .01$) in Group III (BPA) animals due to excessive oxidative stress. However, administration of fenugreek seeds aqueous extract successfully ameliorated ($p \leq .05$) the inimical outcome of oxidative stress and restored the activity of antioxidant enzymes as shown in Group IV (FSEt + BPA).

3.7 | mRNA expression studies

3.7.1 | Bcl-2

Animals of Group III (BPA) showed notably reduction ($p \leq .001$) in the Bcl-2 levels as compared to Group I (C) animals. Further, FSEt

Treatment groups	Testes weight (in mg)	Sperm concentration ($\times 10^6$ cells/ml)	Sperm motility (%)
Group I (C)	138.5 $\pm$ 10.60	9.42 $\pm$ 0.095	83.75 $\pm$ 3.202
Group II (FSEt)	133.0 $\pm$ 5.737	8.8 $\pm$ 0.82	84.00 $\pm$ 5.29
Group III (BPA)	59.2 $\pm$ 7.182 ^{a1,b1}	2.5 $\pm$ 0.46 ^{a1,b1}	31.50 $\pm$ 1.291 ^{a1,b1}
Group IV (FSEt + BPA)	86.0 $\pm$ 3.651 ^{a1,b1,c2}	6.8 $\pm$ 0.173 ^{a1,b2,c1}	58.25 $\pm$ 1.258 ^{a1,b1,c1}

Note: Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), ¹ $p \leq .001$, ² $p \leq .01$, ³ $p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA).

TABLE 2 Alterations in testis weight, sperm concentration and sperm motility after various treatments

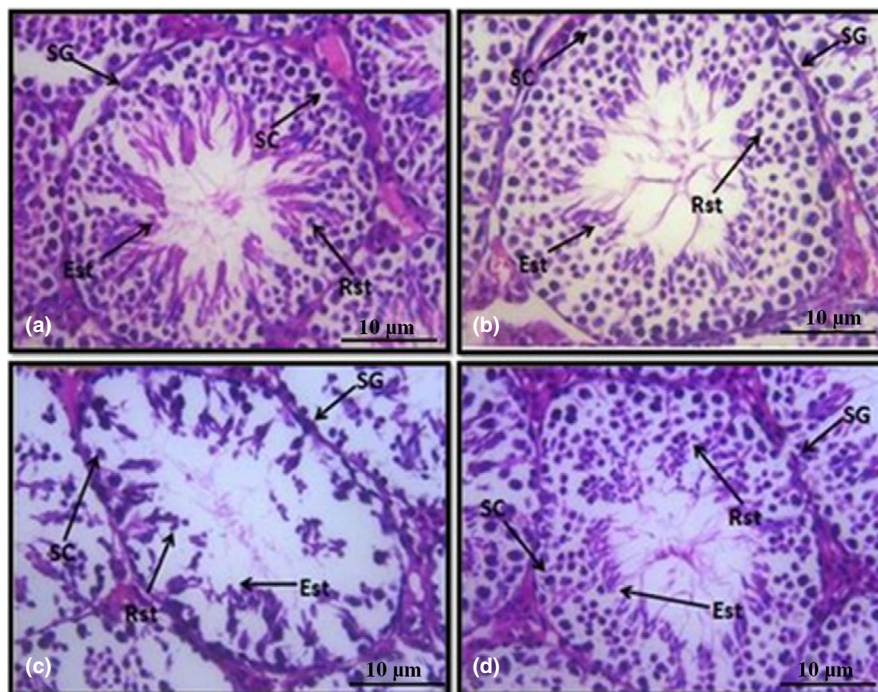


FIGURE 1 Photomicrographs showing H/E staining in paraffin section of mice testes after various treatments in different treatment group at 40X. (a) Group I (C); (b) Group II (FSEt); (c) Group III (BPA); (d) Group IV (FSEt + BPA). Est, Elongated spermatid; Rst, Round spermatid; SC, Primary spermatocytes; SG, Spermatogonia

co-supplementation markedly upregulated ($p \leq .01$) the expression of Bcl-2 in contrast compared with Group III (BPA) (Figure 7).

3.7.2 | Caspase-9

Animals of Group III (BPA) outstandingly increase ($p \leq .05$) in the caspase-9 levels as compared to Group I (C) animals. A significant reduced ($p \leq .05$) expression was observed in Group IV (FSEt + BPA) when compared to Group III (BPA) (Figure 8).

3.7.3 | Caspase-3

Animals of Group III (BPA) impressively increase ($p \leq .001$) the levels of caspase-9 as compared to Group I (C) animals. However, FSEt co-supplementation noticeably reduce ($p \leq .01$) the expression of Bcl-2 when compared with Group III (BPA) (Figure 9).

4 | DISCUSSION

Over the past fifty years, evidences have revealed that the quality of semen has significantly declined worldwide due to the increase in environmental pollution mostly due to industrial factors (Jurewicz, Hanke, Radwan, & Bonde, 2009; Sengupta, Borges, Dutta, & Krajewska-Kulak, 2018). Reproductive system of male has found to be one of the extreme toxicity targets of environmental toxicants. BPA is a well-known environmental toxicant imposing immediate detrimental effect in the testis *via* rupturing the cell junctions between testicular cells therefore resulting increased oxidative stress (Cariati et al., 2019). The antioxidant defence system is reorganised to counteract the adverse effect of increased ROS by reducing the chances of cell death. There are numerous natural antioxidants to scavenge ROS. Fenugreek is one such antioxidant, and aqueous extract of which has been hypothesised to the reduction in lipid peroxidation and enhanced antioxidant status in rats (Hamza et al., 2016).

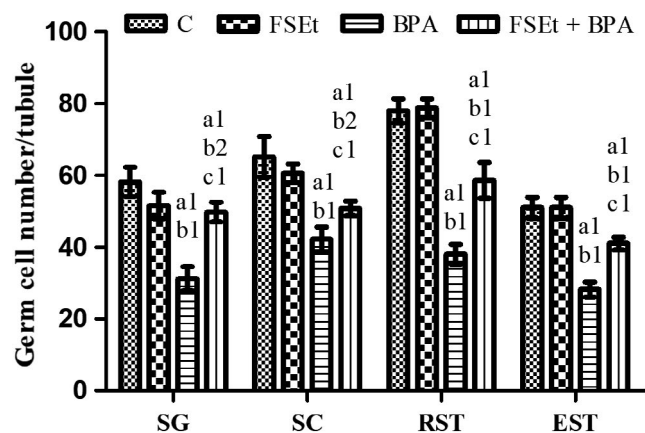


FIGURE 2 Graphical representation of testicular germ cell number in H/E stained paraffin sections of mice testis after a various treatments in different treatment groups. Data are represented as Mean $\pm$ SD of 25 observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

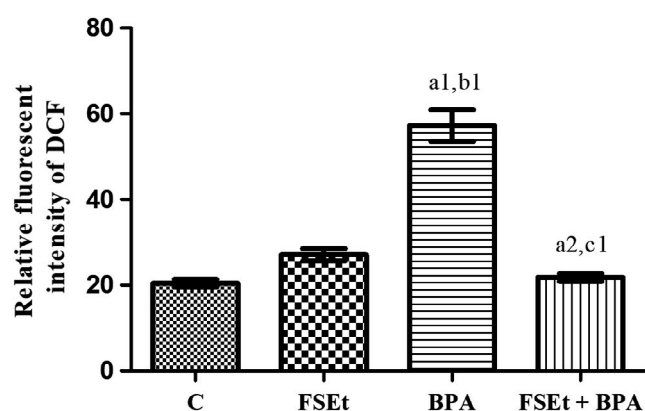


FIGURE 3 Alterations in ROS levels after various treatments. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

Sperm count and motility is one of the main parameters of male infertility. Chemical toxicants exhibit abnormality in spermatozoa by inducing genetic mutation and reproductive toxicity (Cai et al., 2017). Reduction in sperm concentration and motility is an outcome of BPA on the development and growth of spermatogenic cells. These results are in accordance with Wisniewski et al. (2015) where it was reported that BPA significantly damage the acrosomes and the plasma membrane thereby reduces sperm concentration. Spermatogenic cells (spermatogonia, spermatocytes, round spermatids and elongated spermatids) were significantly decreased in mice treated with BPA which might be the reason behind the decline of sperm concentration. Sperm motility and sperm concentration were elevated in

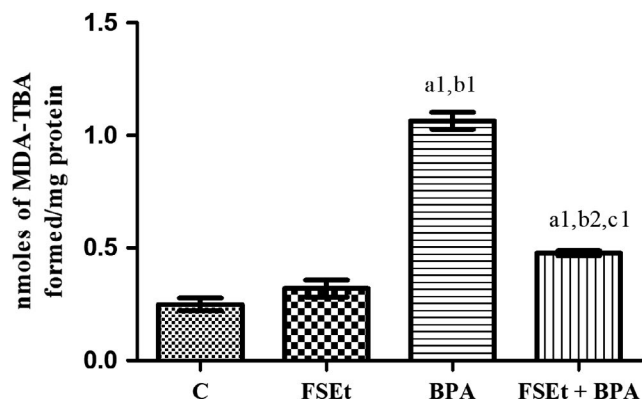


FIGURE 4 Alterations in LPO levels after various treatments. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represent comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

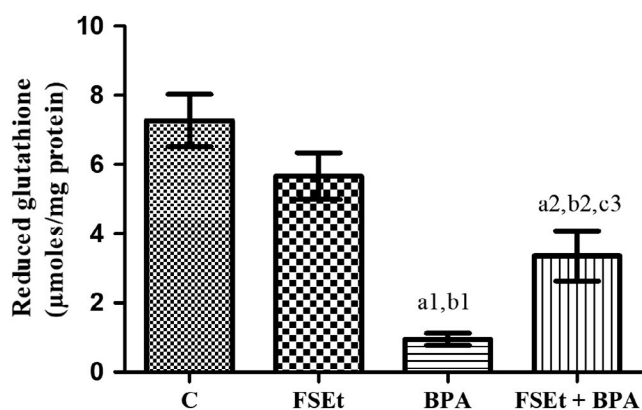


FIGURE 5 Alterations in GSH levels after various treatments. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

mice administered with FSEt demonstrates the antioxidant potential of fenugreek that helps to decrease the oxidative stress.

Further, studies were carried out to evaluate the morphological changes in the testicular cells in different treatment groups. Group I (C) and Group II (FSEt) showed expected testicular histoarchitecture. Additionally, a complete transformation of mature spermatozoa from spermatogonia was clearly observed in the seminiferous tubules. Group III (BPA) showed extensive vacuolisation in germinal epithelium. There was marked reduction in nearly all germ cell population, and the absence of spermatozoa in the lumen was observed. Similar results were detected by Khalaf et al. (2019) where it was reported that BPA exposed group could promote apoptosis of germ cells.

Oxidative stress markers viz ROS and LPO were significantly increased in mice treated with BPA, which may be the possible reason

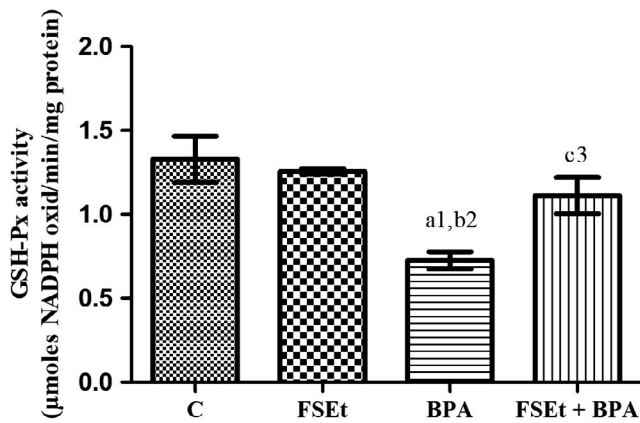


FIGURE 6 Alterations in GSH-Px activity after various treatments. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

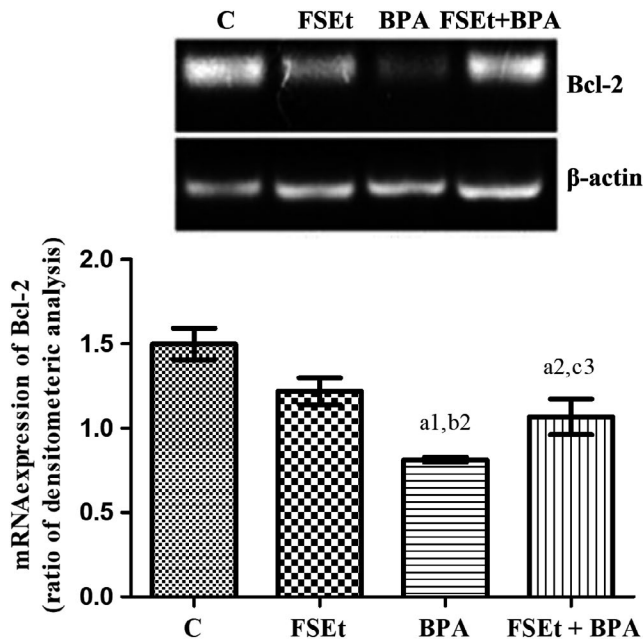


FIGURE 7 Agarose gel expression and densitometric analysis of Bcl-2 mRNA in mice testis by RT-PCR in different treatment groups. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

for the damage of germinal epithelium. Alternatively, systematic production of germ cell was seen in FSEt-administered mice suggesting the ameliorative potential of fenugreek. These results are in accordance with Sakr, El-shenawy, and Al-Shabka (2012) where it was reported that the fenugreek seed extract improved histoarchitecture of the testes and at the same time it reduce the level of serum malondialdehyde and increase the activities of serum superoxide dismutase and catalase.

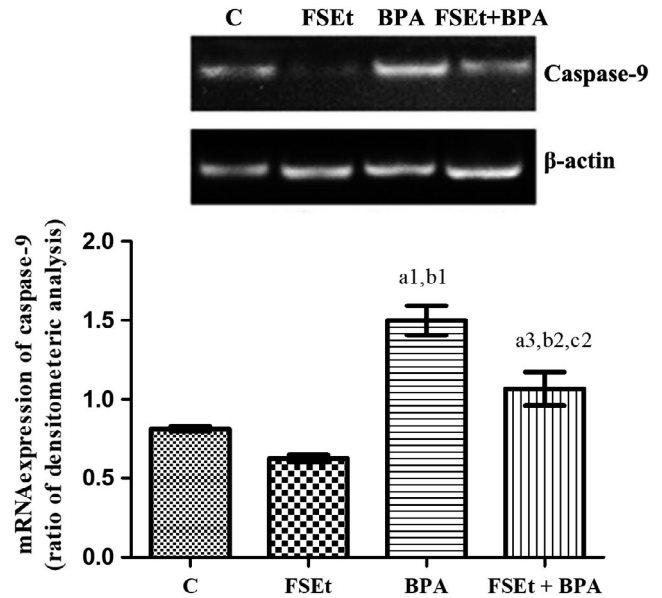


FIGURE 8 Agarose gel expression and densitometric analysis of caspase-9 mRNA in mice testis by RT-PCR in different treatment groups. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

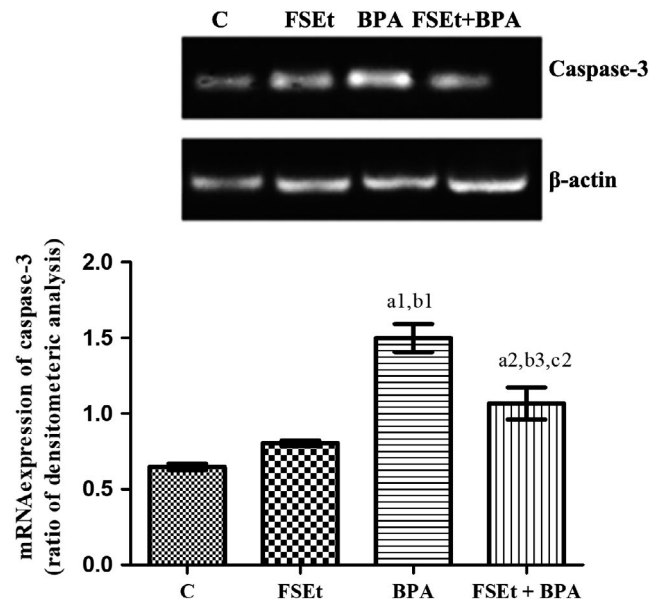


FIGURE 9 Agarose gel expression and densitometric analysis of caspase-3 mRNA in mice testis by RT-PCR in different treatment groups. Data are represented as Mean $\pm$ SD of six observations. Data are analysed by one-way ANOVA followed by post hoc test (LSD), $^1p \leq .001$, $^2p \leq .01$, $^3p \leq .05$. "a" represents comparison w. r. t., Group I (C); "b" represents comparison w. r. t., Group II (FSEt); "c" represents comparison w. r. t., Group III (BPA)

The presence of abundant polyunsaturated fatty acids makes testes to be highly perceptive to oxidative stress. To counteract the adverse effects of ROS, antioxidant defence system is present

that prevent the cellular damage (Prasad, Gupta, & Tyagi, 2017). GSH, and GSH-Px, are antioxidant enzymes that help in the degradation of ROS and thereby reduce oxidative stress. High level of ROS, LPO and low levels of antioxidant enzymes were observed in our study in case of Group III (BPA) which may be due to the accumulation of hydrogen peroxide (H_2O_2) that can lead to oxidative stress (Ullah, Ambreen, Ahsan, & Jahan, 2017). Animals treated with FSEt showed marked improvement in antioxidant enzymes and decrease in the ROS and LPO levels suggests the antioxidant potential of fenugreek. These results are in accordance with the study by Pradeep, and Srinivasan (2017) which suggest that fenugreek decreases the oxidative stress induced in diabetic rats by streptozotocin.

Apoptosis is an essential physiological process and play important roles in diverse physiological processes (Sauler, Bazan, & Lee, 2019). In mRNA studies, decreased levels of Bcl-2 in Group III (BPA) were observed. Diminished expression of Bcl-2 clearly shows the testicular apoptosis. This is in concurrence with the earlier report by Lin et al. (2013) suggesting the decreased expression of anti-apoptotic protein Bcl-2 in dose-dependent manner of BPA exposure. However, there was an increase in the expression of Bcl-2 in Group IV (FSEt + BPA) clearly showing the pivotal role of fenugreek as an antioxidant in the suppression of BPA-induced oxidative stress. Similar results were shown by Khalil, Roshdy, and Kassem (2016) where fenugreek seed extract increased the level of Bcl-2 and showed protective role against Alzheimer's disease.

Caspase activation has been identified as a key mediator of apoptosis. A significant increase in the expression of effector caspase-3 and initiator caspase-9 was perceived in Group III (BPA). These results indicate the onset of apoptotic condition in germ cells and also show a positive relationship between sperm damage by ROS and high levels of caspases. This is also in accordance with Su et al. (2018) where increase expression of caspases was reported in case of BPA treatment. However, decreased level of caspases were observed in FSEt-administered animals, clearly demonstrates the pivotal role of antioxidants present in fenugreek which helps to suppress the BPA-induced oxidative stress and apoptosis.

5 | CONCLUSION

The results reported in this study furnish the evidence that fenugreek modulates the deleterious effects of BPA on the male reproductive system occurring via elevated oxidative stress, and germ cell apoptosis. Thus, the present study pinpoints the new areas of research for the development of better antioxidants for testicular damage.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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How to cite this article: Kaur S, Sadwal S. Studies on the phytomodulatory potential of fenugreek (*Trigonella foenum-graecum*) on bisphenol-A induced testicular damage in mice. *Andrologia*. 2020;52:e13492. <https://doi.org/10.1111/and.13492>